

Table 1: Sequential MAS results for Qwen3-8B. Acc. is accuracy (%). Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS.

Tasks	Metrics	Baselines		LatentMAS variants				Improve
		Single	TextMAS	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting								
AIME24	Acc.	50.0	53.3	56.7	66.25	<u>64.17</u>	63.75	↑ 13.0
	Token	12891	38596	8953	8548	<u>8770</u>	22342	↓ 77.9%
	Time	340.2	1018.6	236.3	225.6	<u>225.8</u>	549.0	× 4.5
AIME25	Acc.	46.7	53.3	53.3	49.58	<u>50.42</u>	49.17	↓ 2.9
	Token	14692	45088	8699	10158	<u>9830</u>	24557	↓ 78.2%
	Time	377.8	1159.4	223.7	261.2	<u>255.5</u>	597.0	× 4.5
GPQA	Acc.	39.9	43.4	45.5	<u>48.74</u>	48.61	49.24	↑ 5.3
	Token	6435	17986	4571	3588	<u>3846</u>	13076	↓ 80.1%
	Time	104.6	292.2	74.3	58.3	<u>59.4</u>	257.6	× 5.0
HumanEval+	Acc.	74.4	80.5	80.5	83.54	<u>83.38</u>	82.62	↑ 3.0
	Token	2507	4593	1866	1637	<u>1696</u>	15055	↓ 64.4%
	Time	31.5	57.7	23.4	18.8	<u>21.5</u>	112.4	× 3.1
MBPP+	Acc.	64.8	69.5	74.6	75.26	<u>75.20</u>	73.15	↑ 5.8
	Token	2053	3695	1164	<u>1462</u>	1528	14388	↓ 60.4%
	Time	28.9	52.1	16.4	<u>17.9</u>	19.6	101.6	× 2.9
MedQA	Acc.	53.0	<u>75.0</u>	75.3	73.67	72.00	66.67	↓ 1.3
	Token	2098	4260	1555	1139	<u>1154</u>	15972	↓ 73.3%
	Time	18.3	37.1	<u>13.5</u>	12.5	13.6	111.1	× 3.0

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 2: Hierarchical MAS results for Qwen3-8B. Acc. is accuracy (%). Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants				Improve
		Single	TextMAS	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>								
AIME24	Acc.	50.0	53.3	53.3	<u>60.00</u>	65.83	—	↑ 12.5
	Token	12891	42629	7526	9926	<u>9898</u>	—	↓ 76.8%
	Time	335.6	1109.7	195.9	258.4	<u>257.9</u>	—	× 4.3
AIME25	Acc.	46.7	50.0	50.0	53.33	<u>52.50</u>	—	↑ 3.3
	Token	14692	53929	13230	11207	<u>11300</u>	—	↓ 79.2%
	Time	379.0	1391.2	341.3	289.1	<u>291.9</u>	—	× 4.8
GPQA	Acc.	39.9	43.0	46.9	51.39	<u>49.75</u>	—	↑ 8.4
	Token	6435	22450	3395	<u>3752</u>	3964	—	↓ 83.3%
	Time	102.7	358.4	54.2	<u>59.9</u>	61.5	—	× 6.0
HumanEval+	Acc.	74.4	76.8	78.0	83.84	<u>83.23</u>	—	↑ 7.0
	Token	2507	8768	1274	1681	<u>1637</u>	—	↓ 81.3%
	Time	30.7	107.5	15.6	<u>20.1</u>	21.0	—	× 5.3
MBPP+	Acc.	64.8	71.9	72.2	<u>72.75</u>	72.16	73.21	↑ 0.8
	Token	2053	7703	1264	1502	<u>1460</u>	12768	↓ 81.0%
	Time	33.8	126.8	20.8	20.1	<u>20.2</u>	92.2	× 6.3
MedQA	Acc.	53.0	<u>76.3</u>	77.0	73.33	72.67	—	↓ 3.0
	Token	2098	6893	1007	1351	<u>1336</u>	—	↓ 80.6%
	Time	25.1	82.5	12.1	15.7	<u>14.8</u>	—	× 5.6

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 3: Sequential MAS results for Qwen3-14B. Acc. is accuracy (%). Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants				Improve
		Single	TextMAS	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting								
AIME24	Acc.	63.3	63.3	66.7	72.08	<u>69.58</u>	—	↑ 8.8
	Token	11263	32092	10593	8259	<u>8489</u>	—	↓ 74.3%
	Time	348.6	993.2	327.8	255.6	<u>283.4</u>	—	× 3.9
AIME25	Acc.	56.7	<u>60.0</u>	63.3	56.67	<u>60.00</u>	49.58	↑ 0.0
	Token	11298	44618	11402	10266	<u>10402</u>	21403	↓ 77.0%
	Time	369.2	1458.1	372.6	335.5	<u>350.6</u>	575.5	× 4.3
GPQA	Acc.	48.5	51.5	52.0	55.81	<u>54.42</u>	53.54	↑ 4.3
	Token	5547	12676	5454	<u>3009</u>	2995	18528	↓ 76.4%
	Time	109.5	250.2	107.7	59.4	<u>67.2</u>	222.9	× 4.2
HumanEval+	Acc.	76.8	81.1	86.5	<u>88.11</u>	86.43	90.70	↑ 7.0
	Token	2366	5934	2042	1594	<u>1607</u>	19587	↓ 73.1%
	Time	31.8	79.7	<u>27.4</u>	24.1	28.1	157.3	× 3.3
MBPP+	Acc.	68.5	72.8	75.7	<u>78.11</u>	79.43	—	↑ 6.6
	Token	1858	4971	1621	1336	<u>1346</u>	—	↓ 73.1%
	Time	28.5	76.2	<u>24.9</u>	21.5	25.5	—	× 3.5
MedQA	Acc.	64.7	80.3	<u>80.7</u>	81.67	79.33	79.00	↑ 1.4
	Token	1746	3444	1841	<u>922</u>	912	22875	↓ 73.5%
	Time	21.1	41.6	22.2	13.2	<u>14.3</u>	151.0	× 3.2

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 4: Hierarchical MAS results for Qwen3-14B. Acc. is accuracy (%). Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants				Improve
		Single	TextMAS	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>								
AIME24	Acc.	63.3	70.0	73.3	69.17	69.58	<u>71.25</u>	↓ 0.4
	Token	11263	29025	10230	8947	<u>9017</u>	23924	↓ 69.2%
	Time	357.6	921.7	324.8	284.1	<u>291.0</u>	639.9	× 3.2
AIME25	Acc.	56.7	66.7	66.7	<u>59.17</u>	57.92	—	↓ 7.5
	Token	11298	50003	9527	<u>10755</u>	11122	—	↓ 78.5%
	Time	364.4	1612.8	307.3	<u>346.9</u>	365.7	—	× 4.6
GPQA	Acc.	48.5	52.0	53.0	<u>54.80</u>	55.81	54.55	↑ 3.8
	Token	5547	20931	3606	<u>2769</u>	2755	20909	↓ 86.8%
	Time	123.2	464.9	80.1	61.5	<u>63.6</u>	287.8	× 7.6
HumanEval+	Acc.	76.8	84.1	86.6	<u>87.04</u>	86.74	91.16	↑ 2.9
	Token	2366	8114	1512	<u>1454</u>	1430	21403	↓ 82.4%
	Time	26.4	90.6	16.9	<u>24.0</u>	25.4	185.6	× 3.8
MBPP+	Acc.	68.5	73.0	73.8	<u>78.04</u>	78.31	—	↑ 5.3
	Token	1858	7458	1187	1132	<u>1182</u>	—	↓ 84.8%
	Time	<u>16.3</u>	65.4	10.4	21.6	23.7	—	× 3.0
MedQA	Acc.	64.7	78.0	<u>78.3</u>	79.00	77.67	48.33	↑ 1.0
	Token	1746	5473	899	860	<u>875</u>	23700	↓ 84.3%
	Time	31.6	99.0	16.3	12.4	<u>13.8</u>	173.7	× 8.0

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.